

Current Stabilization Recovery

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 Prototype Recovery Report.pdf 4 MB

Hardware Report: Prototype Current Stabilization Circuit

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 20220318_Report_Current stabilization project.zip 42 MB

Checking if Y works

heating up is an issue: working with square wave 770ms period 0V to 272mV this corresponds to $V=IR$
 $R=3.3k$ $I = 0.0000824A$, so that means the current in the "coil" is 20mA. after 20s smelling smoke and turned off Test Y: check connections again Y works response time 16ms on 1V and only P of 0.7e0

y z and mot works well, with following settings 6/22/2026

Setpoint: 1khz 405mv high 0v low duty 50%

Current BOM:

IXTK60N50L2 mosfet

AUIRGPS4070D0 IGBT

LEM ITN 12-P ULTRASLAB current transducer

100 ohm sensing resistor

3.3k flyback resistors

5KP58A TVS diodes

V150LA20AP varistor

EVAL-ADUM4221-1EBZ IGBT driver

to be added

47 ohm gate resistor

BUF634AIDR voltage buffer for mosfet gate

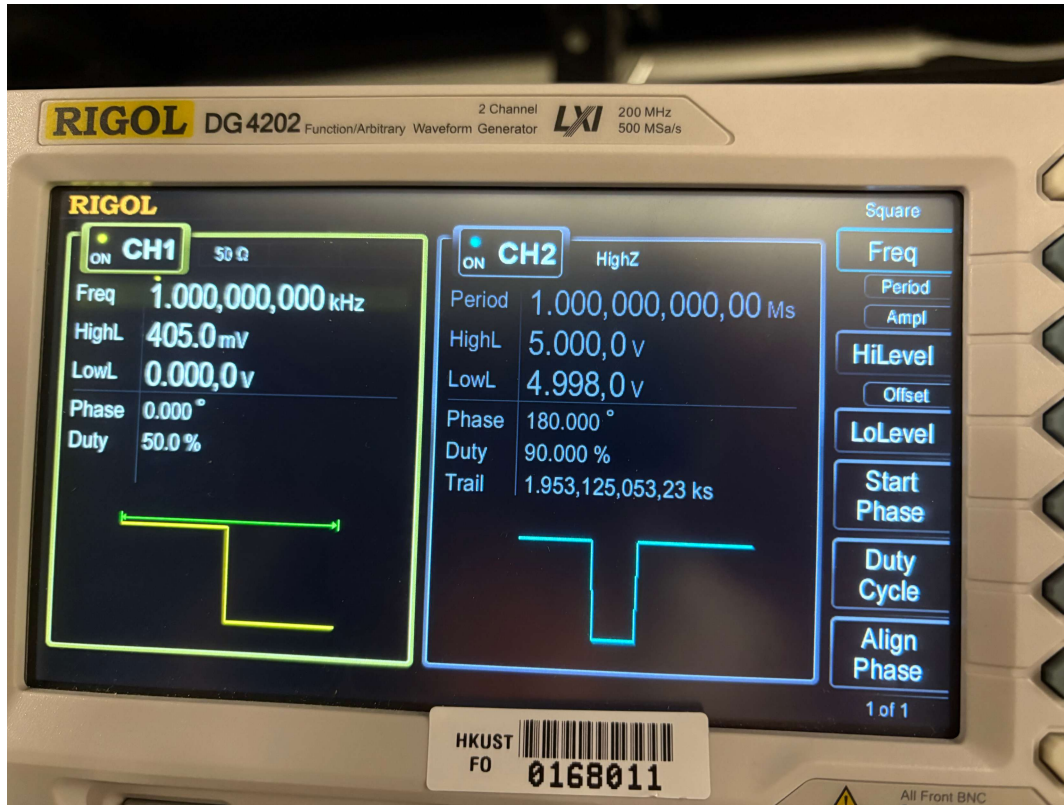
New H-bridge

Pololu G2 High-Power Motor Driver 24v21

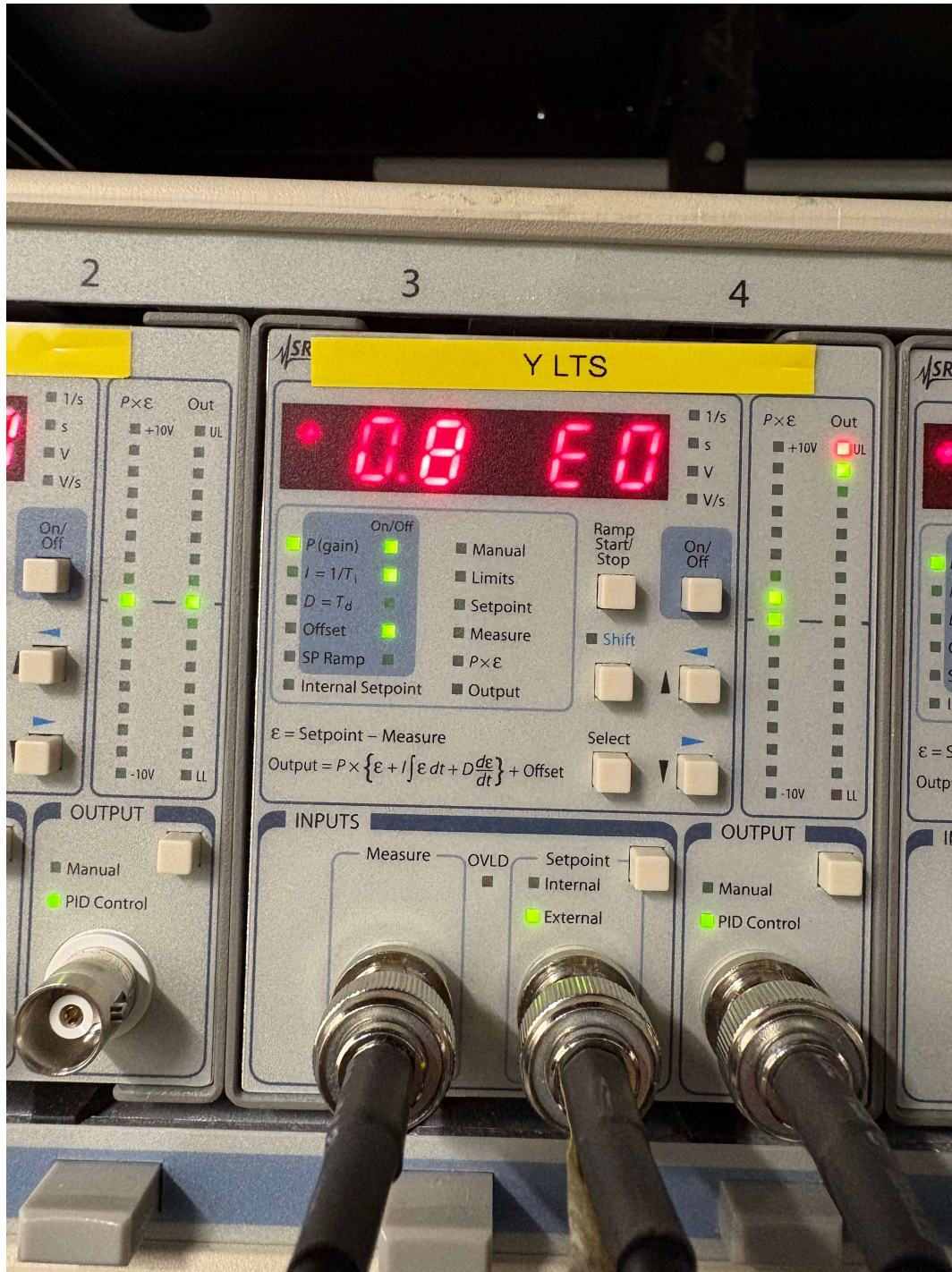
Sharp Transistor Output Optocouplers

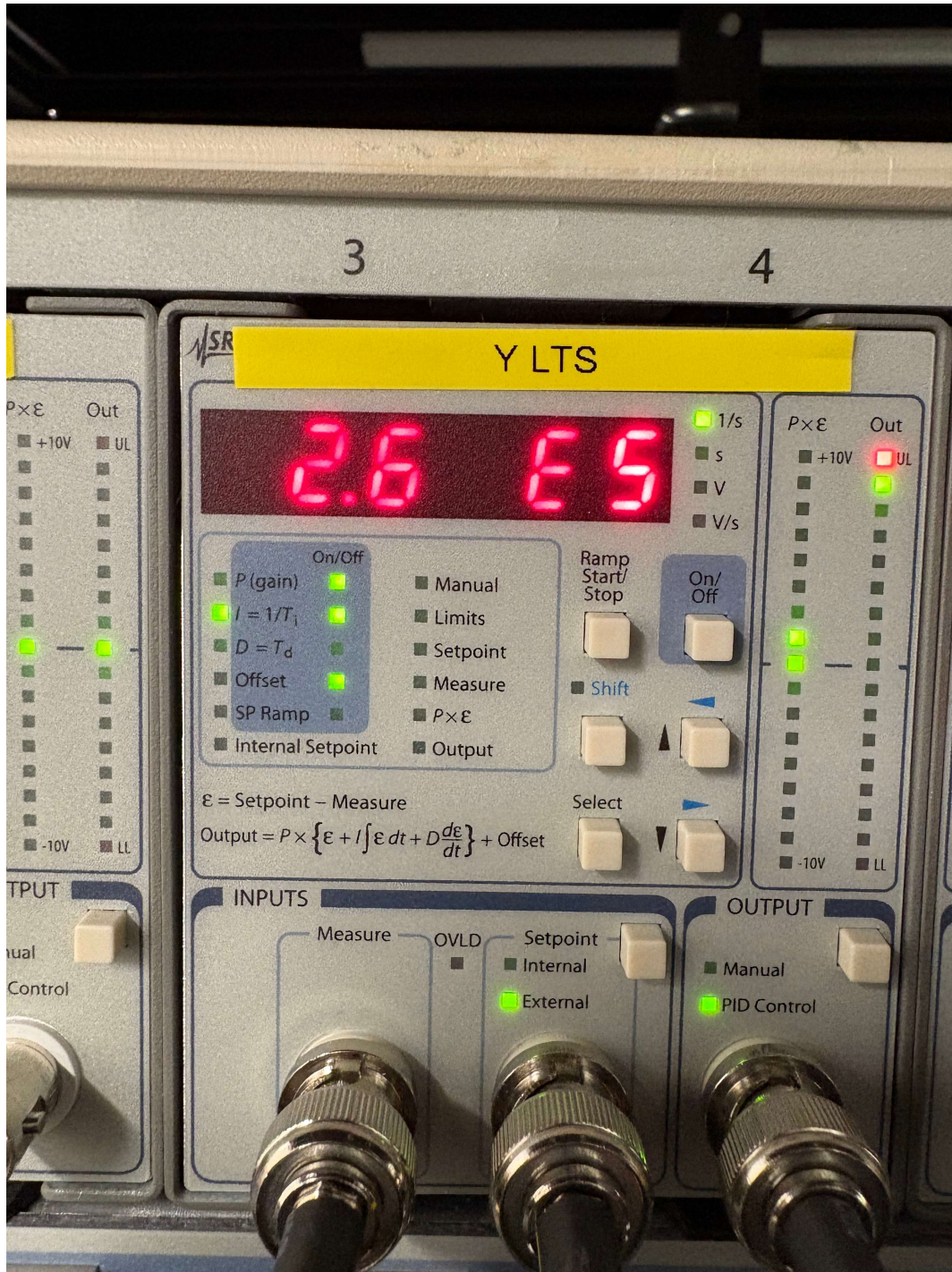
Diotec Semiconductor ESD Protection Diodes / TVS Diodes same TVS diodes

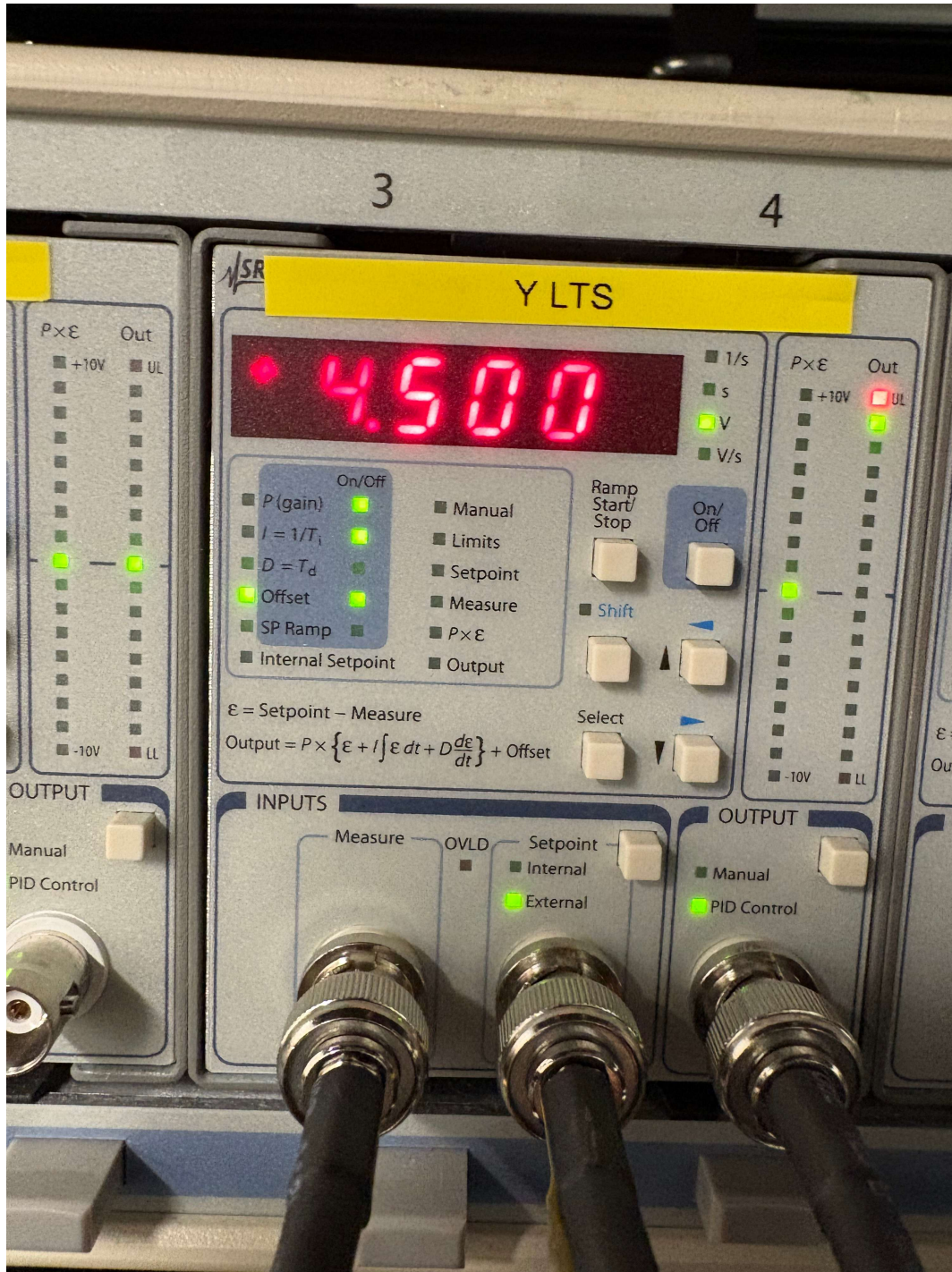
2200uF 50 Volts 20% Nichicon Aluminum Electrolytic Capacitors - Radial Leaded



CH1 is the setpoint, ch2 is pwm







Results:

MOT



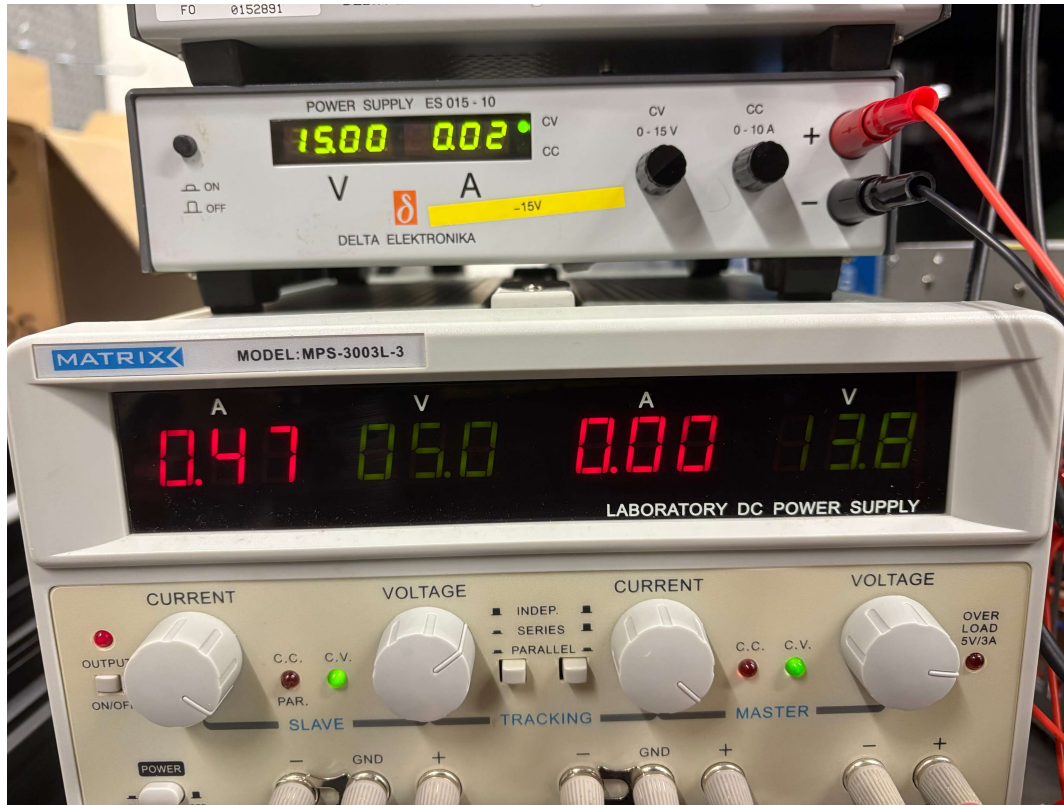
Z



Y



In the X-axis, the IGBT driver is broken, as there seems to be a short circuit



it is sucking up 0.47A

Response time:

Somehow at setting of 400mV in the function generator, the oscilloscope and target shows 800mV.

400mV = 1A

800mV = 2A

response time are all approximately 30us rise time. compared to the thesis value of 500us for 1A and 2A. this might be because there is no coil connected. it could be improved even more by tuning the PID

What if suddenly increasing/decreasing or modulating the current as a step-wise function (a square waveform)? In Fig. 4.8, we have presented the response of the feedback

47

circuit with a step-wise function as an input. It takes less than $50 \mu\text{s}$ to attain the maximum value of the applied current. The current drops suddenly to the minimum value. However, the sudden drop of current introduces overshooting, which is approximately 10% of the maximum value. Overshooting may be troublesome while working near Feshbach resonances. This can be further improved by fine-tuning the PID parameters.



Figure 4.8: Modulating 5.5 A of current with H-bridge as step-wise function. The left image is for 3 kHz and the right image is for 5 kHz oscillating frequency.

actually they do have the 30us

Next step:

1. buy the <https://www.analog.com/en/resources/evaluation-hardware-and-software/evaluation-boards-kits/eval-adum4221-1.html?doc=eval-adum4221-1ebz-ug-1857.pdf> EVAL-ADUM4221-1 \$67 check
2. tune PID parameters even better check
3. test with coil
4. draw PCB for a replica setup
5. Test the replica
6. maybe try some new diodes no we go low channel
7. setup cicero

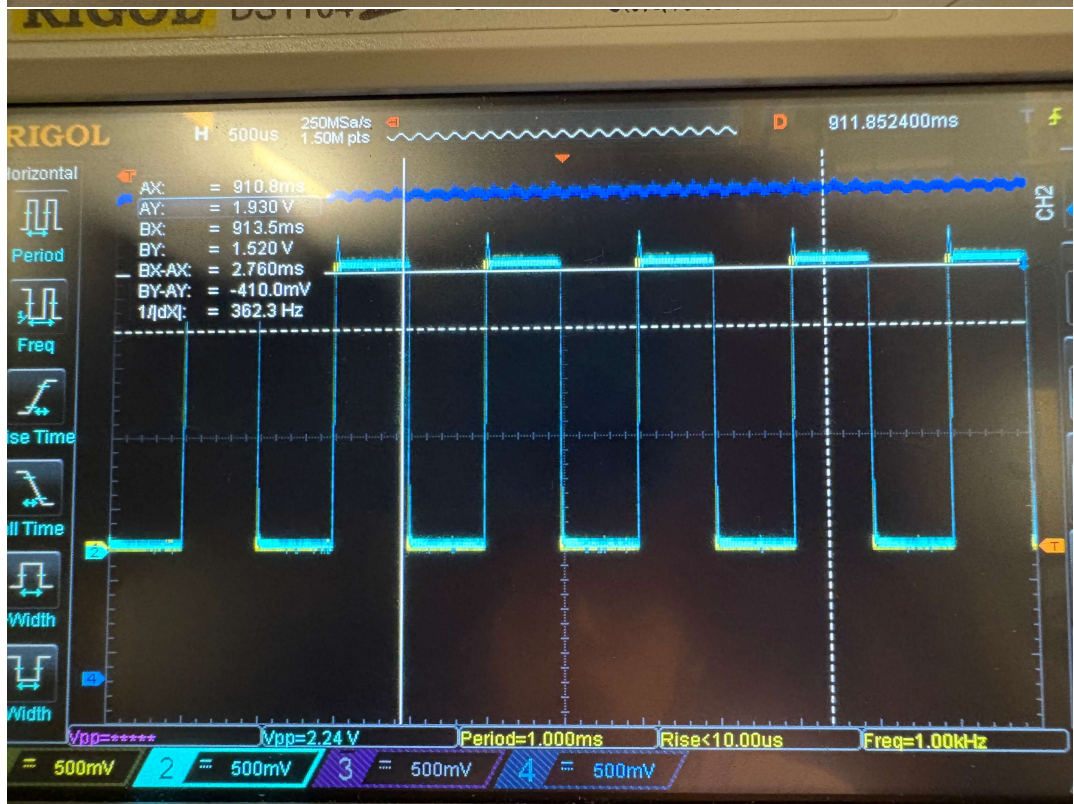
When these are not possible, test the H-bridge

I optimized pid now, the rise time is 26us on Y. changing the power supply voltage from 18V to 30V yielded very little difference in rise time.

also find out how to make a better system with more manageable p and i

replicate the experiments

5A ramping up and down experiments





notice the slight overshooting in square wave

Pololu G2 High-Power Motor Driver 24v21

maybe good hbirdge

replicate the triangular and square experiments with no zero crossing for now

replicated, there is obvious phase lag. The phase lag and the oscillations can be explained by coming near the RC limit of the mosfet gate. The capacitance of the gate is hard to be driven by the analog PID itself, we need a buffer. Then we can rely more on the P and not the I more.

also h brdige like this

647-UHW1H222MHD, quantity 10

852-PC81712NSZ1B, quantity 10

637-5KP58A, quantity 10

595-BUF634AIDR, quantity 10

1. The High-Power Path (30V & Ground)

This is the main muscle of the circuit where the 10A current flows.

- **Power Source (+30V):** Connects directly to the **Primary Input** of your Current Transducer.
- **Current Transducer (Primary Output):** Connects to the **VM (Motor Power)** terminal on the Pololu G2 H-Bridge.
- **Pololu H-Bridge (OUTA and OUTB):** Connect these two terminals directly to the two leads of your Helmholtz Coil.
- **Pololu H-Bridge (GND):** Connects to the **Drain** pin of your custom N-Channel MOSFET.
- **MOSFET (Source):** Connects directly to **System Ground (0V)**.

2. The Mandatory Protection Components

These sit locally on the H-Bridge to catch the inductive kickback during your rapid polarity flips.

- **2200µF Low-ESR Capacitor:** * Positive lead connects to Pololu **VM**.
 - Negative lead connects to Pololu **GND** (which is also the MOSFET Drain node).
- **33V TVS Diode:** Connects directly in parallel with the capacitor, across Pololu **VM** and **GND**.

3. The Current Sensing & PID Loop (The Brain)

This translates the 10A current into a low-voltage signal to command your MOSFET.

- **Current Transducer (Secondary Output):** Connects to one end of your **100-ohm burden resistor**.
- **100-ohm Resistor (Bottom):** Connects to **System Ground**.
- **PID Controller Input:** Taps the line exactly *between* the Transducer Secondary Output and the 100-ohm resistor (yielding your 4V signal at 10A).
- **PID Controller Output:** Connects to the input of your BUF634A gate driver.

4. The BUF634A Gate Drive (The Muscle for the MOSFET)

This powers the MOSFET gate cleanly based on the PID's analog command.

- **BUF634A Pin 7 (V+):** Connects to a dedicated, clean **+12V to +15V Power Supply** (Do *not* use the 30V supply).
- **BUF634A Pin 4 (V-):** Connects to **System Ground**.
- **Buffer Decoupling:** Connect a **0.1µF** and a **10µF** capacitor in parallel directly across Pin 7 and Pin 4 of the BUF634A.
- **BUF634A Pin 3 (Input):** Connects directly to the **PID Controller Output**.
- **BUF634A Pin 6 (Output):** Connects to one side of your **10-ohm to 47-ohm Gate Resistor**.
- **Gate Resistor:** The other side connects directly to the **MOSFET Gate**.
- **Safety Pull-down:** Connect your **3.3k-ohm Resistor** directly between the **MOSFET Gate** and **System Ground**.

5. The NI DAQ & Polarity Control (The Switcher)

This strictly commands the Pololu H-bridge to hold or flip the magnetic field.

- **NI DAQ Digital Out 1 (Enable):** Connects to Channel 1 Input of the **Optoisolator Board**.
- **NI DAQ Digital Out 2 (Direction):** Connects to Channel 2 Input of the **Optoisolator Board**.
- **NI DAQ Ground:** Connects to the input-side Ground of the Optoisolator. (*Do not connect this to your main 30V System Ground. Keep it isolated*).
- **Optoisolator Output 1:** Connects to the Pololu **PWM** logic pin (Set this high to keep the H-bridge 100% ON).

- **Optoisolator Output 2:** Connects to the Pololu **DIR** logic pin (Toggle this to flip the 10A polarity).
- **Optoisolator Output Ground:** Connects to the Pololu logic **GND** pin (which ultimately ties back to System Ground).

Summary of How the Circuit Operates Now

Your 30V supply pushes current through the transducer, into the H-bridge, through the coil, out of the H-bridge, and down into the MOSFET. The NI DAQ strictly tells the H-bridge which direction the current should flow through the coil. Meanwhile, the transducer constantly monitors the total current, telling the PID controller how much to open or close the MOSFET at the bottom of the circuit to hold the current exactly at 10A.

maybe change to low side mosfet instead of high side. this way V_{gs} is not floating on source voltage, and we can introduce the buffer between the mosfet gate and the pid controller. We also eliminate the IGBT, as they are basically useless.

Test without igbt

at 1A, voltage drop across IGBT is 0.6V, voltage drop across the mosfet is 17V insane heat 17watts

similarly at 1A square wave, the fall time with mosfet input gate signal change is 25.8us

I tried using PWM or DISABLE at the current stabilization circuit, but nothing happens. I see the jumpers, and I wonder if the IGBT is ever used at all.

IM06B20AC1

consider this IPM to act as a h-bridge

SRS SIm960 max 20mA

Present:

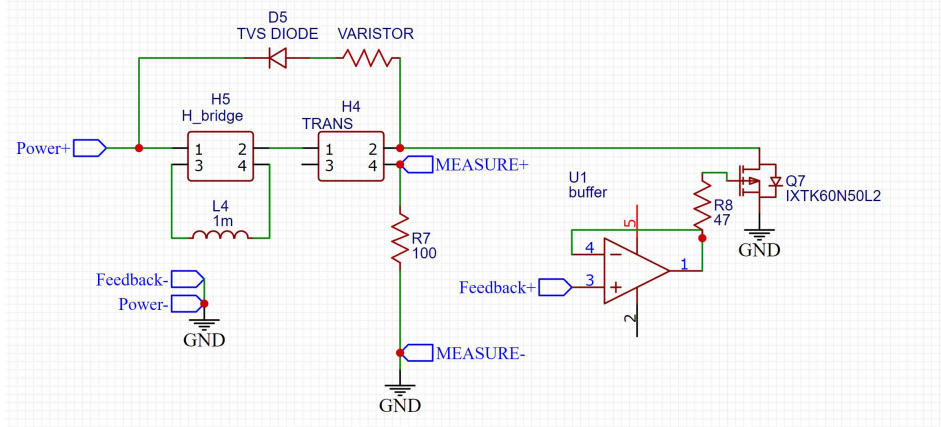
1. Tested coil for shorts and connectivity. No short, only Z bias is disconnected
2. Current Stabilization Circuit figured out
 - a. Y,Z,MOT working as expected
 - b. X-axis has broken IGBT driver. We know because it is taking 0.47A
 - c. Circuit requires low P and high I to function, with only P it reaches 30% of target, the required I is almost at upper limit
 - d. Replication of results
 - i. Rise time ~30us for all axes for a step function. Same as the thesis
 - ii. Fall time also ~30us
 - iii. Phase shift near 5kHz
 - e. Phenomena observed:
 - i. Rapid oscillation when reaching 30% amplitude using only P, very unstable circuit
 - ii. Phase shift with current PID settings near 5kHz
 - iii. Explained by directly driving the mosfet gate with SIM960 PID

1. Max output current 20mA
2. Using a buffer amplifier at gate, we can drive the mosfet gate's capacitance much faster eliminating phase shift
3. The mosfet gate charge is 610nC, PID takes 24us to charge the gate. With the buffer it will be 10 times faster at under 2.4us
4. With the buffer we can also tune to a more aggressive P and lower I, as I effective is low pass filter that delays the rise signal.

f. Problems with the circuit

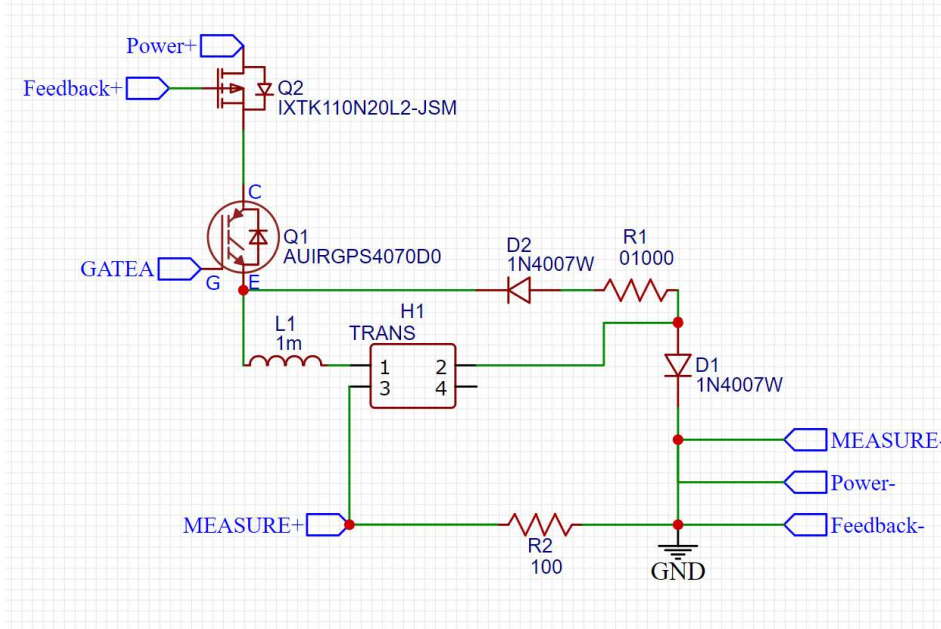
i. To add buffer we need to shift to a low-side mosfet design

ii.

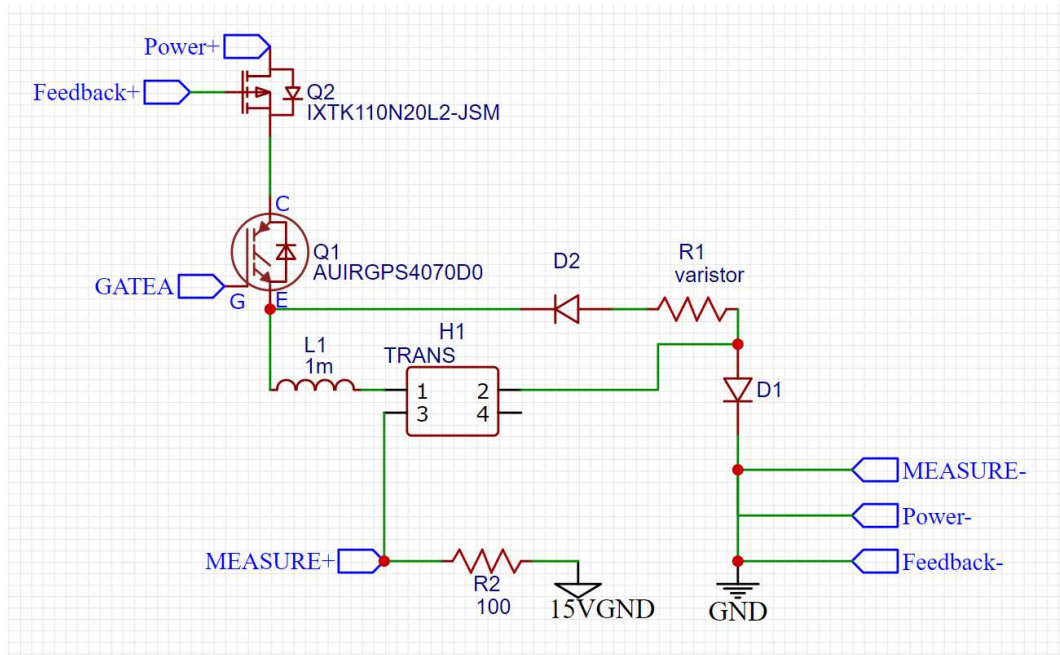


New design

iii.



Old design



- i.
 - ii. This is because the buffer needs to connect to ground directly, not a floating source in the mosfet
 - iii. The D1 diode doesn't appear in the thesis, and normal flyback diodes do not have this. If we change to low-side mosfet then it has no where to be. Mithilesh not reply yet
 - iv. The IGBT in the current stabilization circuit seems to have no use. No experimental sequence is told to use the IGBT as the switch.
 - 1. In theory, when PWM is low, the switch is off, and when DISABLE is high, the switch should be off no matter what
 - 2. Tried giving logic to turn off, but the driver is providing gate voltage no matter what
 - 3. On the board there are jumpers that physically wire these pins high or low, but still haven't figured out. Mithilesh not reply yet
 - 4. Shutting down the mosfet should be just as fast as shutting down with IGBT, as the fall time is limited by the rate of power dissipation by the varistor
 - 5. Power is also wasted, as the on IGBT has 0.6V across it, wasting 0.6W at 1A and 6W and 10A and so on
1. H-bridge new design
- a. We need ability to withstand 500V instantaneous voltage difference, 10A continuous, and low heating. Power supply at most 30V, the varistor max voltage 264V, sum to around 300V
 - b. Use commercial Intelligent Power Module
 - i. used in three-phase electricity, we just use 2 phases to use as H-bridge
 - ii.



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Product details

[Parameters](#) [Ordering](#) [Features](#) [Benefits](#) [Applications](#)

Additional Information		Electrical	Product Category	Intelligent Power Modules (IPM)
Built in NTC	Yes		Product Group	CIPOS™ Mini
Configuration	3 Phase Open Emitter		Product Name	IM06B20AC1
Encryption	No		Qualification	Industrial
I _c (@ 25°)	20 A		Rated Current	20 A
Language	PLECS		Switch Type	IGBT
P _{max} (10kHz)	2400 W		Voltage Class	600 V

iii. Already order 5 samples. Need to draw custom PCB to mount it. It requires very little work, no protection and no capacitors, but we should still use some

2. Old H-bridge recovery

- Some broken cables, still trying to put it together, the two designs are different with completely different labeled cables, very confusing to recover.

Next steps:

Draw PCB for IPM, fix old H-bridge, Assemble new h-bridge, test the current stabilization with low-side design and the buffer.

Work this week

Monday: Draw IPM PCB with DCDC converter, order DCDC converter, order PCB

Tuesday: If the driver comes, take apart X axis and assemble low side with the driver and test

Wednesday, If the driver and low side work, start drawing finalized PCB for the full design, also do some calculation on voltage required

Thursday: Whenever the DCDC and PCB comes, assemble and test it

Friday: same as thursday, also whatever comes up maybe test the TDK lambda

need to buy the LEM, the current sense resistor, the mosfet.

Product Index > Resistors > Through Hole Resistors > Vishay Beyschlag/Draloric/BC Components MBE04140C1000FC100

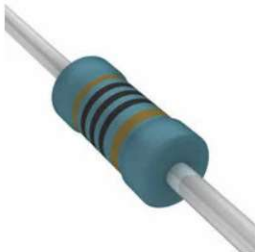


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

MBE04140C1000FC100

DigiKey Part Number

BC4517CT-ND - Cut Tape (CT)
BC4517TB-ND - Tape & Box (TB)

Manufacturer

Vishay Beyschlag/Draloric/BC Components

Manufacturer Product Number

MBE04140C1000FC100

Description

RES 100 OHM 1% 1W AXIAL

Manufacturer Standard Lead Time

13 Weeks

Customer Reference

Detailed Description

100 Ohms ±1% 1W Through Hole Resistor Axial Automotive AEC-Q200 Metal Film

Datasheet

 [Datasheet](#)

Product Attributes

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Category	Resistors Through Hole Resistors	Composition	Metal Film
Mfr	Vishay Beyschlag/Draloric/BC Components	Features	Automotive AEC-Q200
Series	MBE/SMA 0414 - Professional	Temperature Coefficient	±50ppm/°C
Packaging	Cut Tape (CT) ⓘ Tape & Box (TB) ⓘ	Operating Temperature	-55°C ~ 155°C
Part Status	Active	Package / Case	Axial
Resistance	100 Ohms	Supplier Device Package	Axial
Tolerance	±1%	Size / Dimension	0.165" Dia x 0.469" L (4.20mm x 11.90mm)
Power (Watts)	1W	Number of Terminations	2

Report Product Information Error

Product Index > Resistors > Through Hole Resistors > Vishay Dale.CMF55100R00BEEB



Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

CMF55100R00BEEB

DigiKey Part Number

CMF100HBTR-ND - Tape & Reel (TR)
CMF100HBCT-ND - Cut Tape (CT)

Manufacturer

Vishay Dale

Manufacturer Product Number

CMF55100R00BEEB

Description

RES 100 OHM 1/2W 0.1% AXIAL

Manufacturer Standard Lead Time

10 Weeks

Customer Reference

Detailed Description

100 Ohms $\pm 0.1\%$ 0.6W Through Hole Resistor Axial Flame Retardant Coating, Moisture Resistant, Safety Metal Film

Datasheet

 [Datasheet](#)

EDA/CAD Models

[CMF55100R00BEEB Models](#)

Product Attributes

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Category	Resistors Through Hole Resistors	Features	Flame Retardant Coating, Moisture Resistant, Safety
Mfr	Vishay Dale	Temperature Coefficient	± 25 ppm/ $^{\circ}$ C
Series	CMF	Operating Temperature	-55 $^{\circ}$ C ~ 175 $^{\circ}$ C
Packaging	Tape & Reel (TR) ⓘ Cut Tape (CT) ⓘ	Package / Case	Axial
Part Status	Active	Supplier Device Package	Axial
Resistance	100 Ohms	Size / Dimension	0.090" Dia x 0.240" L (2.29mm x 6.10mm)
Tolerance	$\pm 0.1\%$	Number of Terminations	2
Power (Watts)	0.6W	Base Product Number	CMF55
Composition	Metal Film		

Report Product Information Error

To wire the LT1210 as a stable buffer, you must:

1. Connect your SRS PID signal to the **Non-Inverting Input (+)**.
2. Place a **Feedback Resistor (\$R_F\$)** between the **Output** and the **Inverting Input (-)**. (Check the LT1210 datasheet tables, but for a gain of +1, they usually recommend an \$R_F\$ of about $\$510\Omega$ to $\$1\text{k}\Omega$).
3. Connect the Output through a small $\$10\Omega$ **Gate Resistor** directly to your MOSFET Gate.

Item,Qty,Designator,Part / Value,Description & Specifics

1,1,U1,LT1210,"Current Feedback Amplifier. At $\pm 15\text{V}$, the TO-220 (LT1210CT7) or DD (LT1210CR) package is highly recommended over the SO-16 to handle thermal dissipation. PDF"

2,1,Q1,User's MOSFET,The exact low-side MOSFET you are driving.

3,1,RF,750 Ω ,"Feedback resistor connecting OUT to -IN. CRITICAL: Use a 1/4W,1% metal film resistor. Do not use a wire short. PDF"



Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

RN73H2BTTD7500F100

DigiKey Part Number
2019-RN73H2BTTD7500F100TR-ND - Tape & Reel (TR)
2019-RN73H2BTTD7500F100CT-ND - Cut Tape (CT)
2019-RN73H2BTTD7500F100DKR-ND - Digi-Reel®

Manufacturer
[KOA Speer Electronics, Inc.](#)

Manufacturer Product Number
RN73H2BTTD7500F100

Description
RES 750 OHM 1% 1/4W 1206

Manufacturer Standard Lead Time
8 Weeks

Customer Reference

Detailed Description
750 Ohms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) Automotive AEC-Q200, Moisture Resistant Metal Film

Datasheet
 [Datasheet](#)

EDA/CAD Models
[RN73H2BTTD7500F100 Models](#)

Product Attributes

☐ Filter Similar Products ☐ Show Empty Attributes

Category	Resistors Chip Resistor - Surface Mount	Features	Automotive AEC-Q200, Moisture Resistant
Mfr	KOA Speer Electronics, Inc.	Temperature Coefficient	±100ppm/°C
Series	RN73H	Operating Temperature	-55°C ~ 155°C
Packaging	Tape & Reel (TR) ⓘ Cut Tape (CT) ⓘ Digi-Reel® ⓘ	Package / Case	1206 (3216 Metric)
Part Status	Active	Supplier Device Package	1206
Resistance	750 Ohms	Size / Dimension	0.126" L x 0.063" W (3.20mm x 1.60mm)
Tolerance	±1%	Height - Seated (Max)	0.028" (0.70mm)
Power (Watts)	0.25W, 1/4W	Number of Terminations	2
Composition	Metal Film	Base Product Number	RN73H2B

[Report Product Information Error](#)

- 4,1,CCOMP,0.01μF (10 nF),"Ceramic capacitor. Connects between the OUT and COMP pins. This engages the internal compensation network, which is critical for stabilizing the amplifier against the high capacitive load of the MOSFET gate. PDF"
- 5,1,RPD,10kΩ, Gate pull-down resistor. Connects the MOSFET Gate (and LT1210 OUT) to Ground. Prevents the MOSFET gate from floating if the LT1210 is in a high-impedance shutdown state. PDF
- 6,2,"CBYP1,CBYP2",100nF (0.1 μF),High-frequency ceramic bypass capacitors (Rated for 25V or 50V). Must be placed as close to the LT1210's V+ and V- pins as physically possible. PDF
- 7,2,"CBULK1,CBULK2",4.7μF,"Low-frequency bulk bypass capacitors (Solid Tantalum or Ceramic, Rated 25V or higher). Place immediately next to the 100nF capacitors on both the V+ and V- rails. PDF+ 1"

how many units
how many amps